

**ADMINISTRATIVE INFORMATION****FMI No: 25383****FMI Title:** Patch A for VCT M4 06MW03.4 SW RIs

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support		
GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems On-Line Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

This FMI is for 7.X LightSpeed 32 & 64 slice VCT systems with 06MW03.4 Q2 2006 software.

The gantry model numbers tracked are: 5124069 and 5126093.

Unit Tracked: 5124069 and 5126093**FMI Instructions Part No.:** 5178436-100**Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

GEMS - Europe

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25383

FMI Title: Patch A for VCT M4 06MW03.4 SW RIs

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Effectivity

This FMI is for 7.X LightSpeed 32 & 64 slice VCT systems with 06MW03.4 Q2 2006 software.

The gantry model numbers tracked are: 5124069 and 5126093.

Purpose

The purpose of this FMI is to load Patch A for VCT M4 software release 06MW03.4. This patch provides the quality fixes outlined below:

- Dark or distorted images
 - o Images may appear dark or distorted during initial reconstruction, but recon correctly if retrospective reconstruction is completed.
 - o This issue was found to be caused by error paths in Scan Abort, Stop Scan and Priority Recon logic. This has been corrected.
 - o There is a very small possibility that problem may still exist after patch is loaded, due to a different error path. If so, please write a iTrak Report (PQR) for engineering follow-up.
- Exam Split
 - o Images arrive as unspecified or unverified at the PACS when using Exam Split.
 - o Caused by the Accession number not being populated to the Accession number DICOM field when a new exam was created from Exam Split. The Accession number is now populated correctly.
- Accession Number:
 - o No accession number is listed in the image annotation when more than 1 worklist item has been selected.
 - o This has been corrected. The accession number of the last Work List Item selected in Patient Schedule will be displayed in image annotation.
- Trackball loss of function.
 - o Possibility existed for trackball to lose functionality.
 - o Problem with corrupted configuration file, which has been corrected.

Prerequisites

System must be loaded with software revision 06MW03.4 (VCT M4) prior to loading this patch. If software version 06MW03.4 is not loaded, please upgrade system before installing.

Software revision 06MW03.4 was deployed to the installed base via FMI 25373, kit part number 5145027.

Prerequisite FMIs

FMI Number	Title
25373	VCT M4 Software Release 06MW03.4

Furnished Materials

FMI 25383, Kit Part Number 5178436-2		
Qty	Description	Part #
1	FMI 25383 Document	5178436-100
1	CDROM, Patch A for 06MW03.4 Software	5179392

Personnel Requirements

Travel Time: 1 hr

Labor Time: 1 hr

Procedure

1. Prerequisite:

- The system software must be at 06MW03.4 before installing "Patch A."

To check the software version:

1. Open Unix shell
2. Login as ctuser.
3. Type **swhwinfo -all** <Enter>
4. If "06MW03.4" is part of the returned string, continue with the load of this software patch.

If "06MW03.4" is not part of the returned string, **STOP!**

Update system with 06MW03.4 software (FMI 25373), and then install 06MW03.4's "Patch A."

- If the Patch A has already been installed in your system, **STOP!** This FMI is not needed.

To check whether Patch A has been installed or not,

Type **showprods | grep -i patch** <Enter>

The output will be similar to the following if Patch A has been installed:

```
I VCTM4_patchA <date> Patch A for VCTM4 ver 1 rel 1.1
```

- Please ask your customer which mode of Exam Split should be configured, Hard Exam Split (HES) or Virtual Exam Split (VES). For most (~99%) systems, the option should be set in HES mode. ***If you are not sure, set it in HES mode.***

At this time, only McKesson PACS systems can accept images in VES mode. Please contact Advanced Application Specialist to configure for VES mode.

Refer to Appendix B for an explanation of Exam Split option.

2. Remove Exam Split Option

- Go to Common Service Desktop, select Configuration, select Install Options
- Click on Exam Split Option, then Remove.
- Shutdown and Restart Applications. **You must shutdown and restart Applications to complete removal of Exam Split option.**

3. Installation of "Patch A" using CDROM:

VCT M4 Patch A Installation Instructions

1. Shut down all CT Applications
2. Open Unix Shell
3. Type **su** - <Enter>
4. Type password **#bigguy**
5. Insert 06MW03.4 Patch A CD into the CT host PC's CDROM drive
6. Type **patch_install -c** <Enter>

The output will be as follows:

These patches from cdrom will be installed:

VCTM4_patchA

=====

I will install update VCTM4_patchA from CDROM, is this ok ? [y/n]

7. Type **'y'** <Enter>
8. Confirm that the patch has been successfully installed by typing:

showprods | grep -i patch <Enter>

The output will be similar to the following:

I VCTM4_patchA <date> Patch A for VCTM4 ver 1 rel 1.1

9. Eject CD, type **eject** <Enter>
10. Type **reboot** <Enter>

Wait for applications to start.

3. Re-install Exam Split Option.

- a. Go to Common Service Desktop, select Configuration, select Install Options
Install Exam Split option.

When installing the option, please select the proper mode. See Section 1, Pre-requisites for additional information.



- b. Once the mode is selected and option installed, you will get the following message on the screen:



- c. Shutdown and Restart Applications.

4. Verify Exam Split Mode:

Open a Unix Shell and type the following:

- a. {ctuser@hostname} **su -**
- b. Password: **#bigguy**
- c. [root@hostname] **ls -l ~ctuser/ves/.hesMode**

NOTE There are no spaces in the phrase
: ~ctuser/ves/.hesMode

Examine the results.

- d. If the results are similar to:

```
-rw-r--r-- 1 ctuser users 0 Apr 3 12:43 /usr/g/ctuser/ves/.hesMode
```

Then HES (Hard Exam Split) mode is configured.

If the results show 'No such file or directory', then VES (Virtual Exam Split) mode is configured.

- e. If the results do not show the correct mode (HES or VES), re-install the Exam Split option with the correct mode (HES or VES).

5. Final Checks

Perform sanity scans to ensure proper system performance before returning it to the customer:

- 1. Successfully perform a Scout scan.
- 2. Successfully perform a Helical scan.
- 3. Successfully perform an Axial scan.

6. FMI Closure

Close FMI per regional procedure.

Remove this FMI document from the customer site.

Appendix A

Removal of Patch A

The following instructions are to be used to de-install 06MW03.4 Patch A in the event it must be removed:

Remove / Uninstall 06MW03.4 Patch A:

1. Shutdown all applications
2. Open Unix Shell
3. Type **su** - <Enter>
4. Insert 06MW03.4 PatchA CD into the CT host PC's CDROM drive
5. Type **patch_install -u** <Enter>

You will see the following message:

VCTM4_patchA 1-1.i386 has been uninstalled successfully.

6. Eject CD, type **eject** <Enter>
7. Type **reboot** <Enter>

Wait for applications to start.

Appendix B

Exam Split

This section presents the concept necessary to understand Exam Split (Option). The concept you need to understand is:

- Exam Split
 - Virtual Mode
 - Hard Mode

Exam Split:

Provides you with the capability to "split" a series of patient images into separate groups. These new smaller image groups can be networked to desired reading stations for multiple "reads" and multiple billings on select patient exams.

Using the Exam Split option will allow for split images from a single acquisition and assign them to a Requested Procedure ID or accession number retrospectively. On the ImageWorks desktop using Exam Split, all the images of the scan will be loaded. Using left mouse key to select the first image and shift, left mouse key simultaneously to choose the last image to be sent to a specific exam procedure.

At scan time all Patient records that you wish to have available to split to must be selected from the Patient Schedule when selecting New Patient. Your system will be configured in one of two modes for Exam Split. The mode configured is dependent on the capabilities of the system you are sending images to review.

Exam Split requires that the Connect Pro option be installed.

Virtual Exam Split Mode (VES):

Your remote station must support Performed Procedure Step (PPS) and Gray Scale Presentation State (GSPS). Images will be Auto Transferred to the Remote station. In Exam Split ranges of images will be assigned to each accession number or procedure code and a Gray Scale Presentation State (GSPS) object will be created and transferred when selected.

Hard Exam Split Mode (HES):

Hard Exam Split will create a new series of images for each accession number or procedure code ranges of images are assigned to. For this reason images will not be Auto Transferred to the Remote Station.

The above information is taken from Chapter 13 of the LightSpeed VCT 5119850-247 Learning and Reference Guide.